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Probability and Statistics Lab (DA-1)

Question: For each of the three data sets provided below, calculate the following statistical measures: Mean (using direct method, assumed mean method, step deviation method), Median, Mode, Range, coefficient of range, Quartile Deviation, coefficient of Q.D, Mean Deviation from mean, coefficient of M.D, Mean Deviation from median, coefficient of M.D, Standard Deviation (Direct, Assumed Mean Method, Step Deviation Method).

1. A science teacher administered a chemistry quiz to a class of 25 students. The maximum possible score was 20 points. The teacher recorded the raw scores for each student to analyse the class's performance.

   12, 15, 18, 14, 16, 12, 19, 15, 13, 17, 15, 11, 14, 18, 20, 15, 13, 16, 12, 19, 14, 10, 17, 15, 18

CODE:-

DA1.R*

```r
#Question1:
x <- c(12,15,18,14,16,12,19,15,13,17,15,11,14,18,20,
       15,13,16,12,19,14,10,17,15,18)
#MEAN (Three Methods)
#(a) Direct Method
mean_x <- sum(x) / length(x)
mean_x
#(b) Assumed Mean Method
A <- 15 # assumed mean
d <- x - A
mean_assumed <- A + sum(d) / length(x)
mean_assumed
#(c) Step Deviation Method
h <- 1
u <- (x - A) / h
mean_step <- A + (sum(u) / length(x)) * h
mean_step
#MEDIAN
median_x <- median(x)
median_x
#MODE
freq <- table(x)
mode_x <- as.numeric(names(freq[freq == max(freq)]))
mode_x
#RANGE & COEFFICIENT OF RANGE
range_x <- max(x) - min(x)
range_x
coef_range <- (max(x) - min(x)) / (max(x) + min(x))
coef_range
#QUARTILE DEVIATION & COEFFICIENT
Q1 <- as.numeric(quantile(x, 0.25))
Q3 <- as.numeric(quantile(x, 0.75))
QD <- (Q3 - Q1) / 2
QD
coef_QD <- (Q3 - Q1) / (Q3 + Q1)
coef_QD
#MEAN DEVIATION FROM MEAN & COEFFICIENT
MD_mean <- sum(abs(x - mean_x)) / length(x)
MD_mean
```

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CONSOLE:-

Source

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```
> #Question1:
> x <- c(12,15,18,14,16,12,19,15,13,17,15,11,14,18,20,
+        15,13,16,12,19,14,10,17,15,18)
> #MEAN (Three Methods)
> #(a) Direct Method
> mean_x <- sum(x) / length(x)
> mean_x
[1] 15.12
> #(b) Assumed Mean Method
> A <- 15 # assumed mean
> d <- x - A
> mean_assumed <- A + sum(d) / length(x)
> mean_assumed
[1] 15.12
> #(c) Step Deviation Method
> h <- 1
> u <- (x - A) / h
> mean_step <- A + (sum(u) / length(x)) * h
> mean_step
[1] 15.12
> #MEDIAN
> median_x <- median(x)
> median_x
[1] 15
> #MODE
> freq <- table(x)
> mode_x <- as.numeric(names(freq[freq == max(freq)]))
> mode_x
[1] 15
> #RANGE & COEFFICIENT OF RANGE
> range_x <- max(x) - min(x)
> range_x
[1] 10
> coef_range <- (max(x) - min(x)) / (max(x) + min(x))
> coef_range
[1] 0.3333333
> #QUARTILE DEVIATION & COEFFICIENT
> Q1 <- as.numeric(quantile(x, 0.25))
> Q3 <- as.numeric(quantile(x, 0.75))
> QD <- (Q3 - Q1) / 2
```

Source

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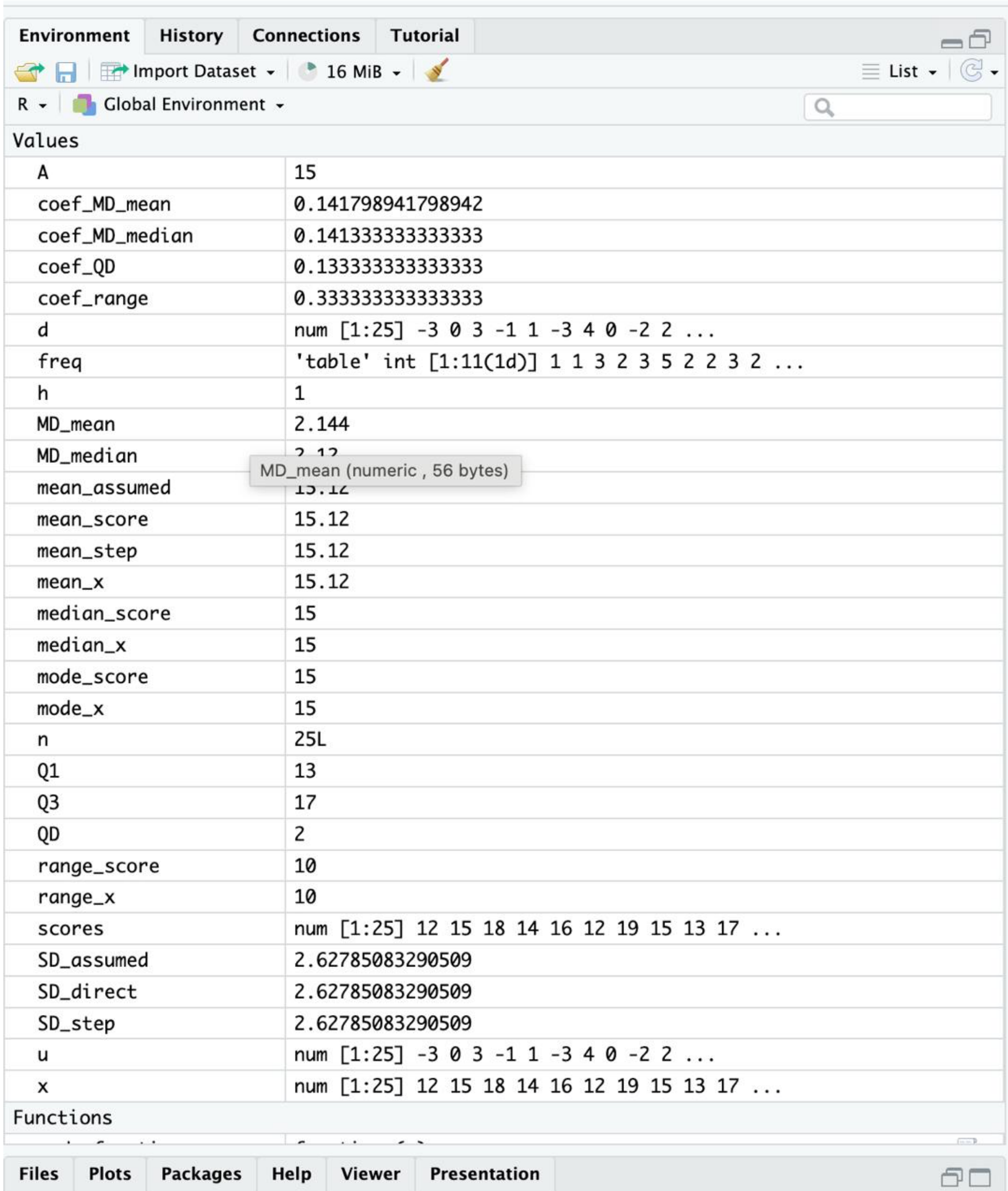
```
> #QUARTILE DEVIATION & COEFFICIENT
> Q1 <- as.numeric(quantile(x, 0.25))
> Q3 <- as.numeric(quantile(x, 0.75))
> QD <- (Q3 - Q1) / 2
> QD
[1] 2
> coef_QD <- (Q3 - Q1) / (Q3 + Q1)
> coef_QD
[1] 0.1333333
> #MEAN DEVIATION FROM MEAN & COEFFICIENT
> MD_mean <- sum(abs(x - mean_x)) / length(x)
> MD_mean
[1] 2.144
> coef_MD_mean <- MD_mean / mean_x
> coef_MD_mean
[1] 0.1417989
> #MEAN DEVIATION FROM MEDIAN & COEFFICIENT
> MD_median <- sum(abs(x - median_x)) / length(x)
> MD_median
[1] 2.12
> coef_MD_median <- MD_median / median_x
> coef_MD_median
[1] 0.1413333
> #STANDARD DEVIATION (Three Methods)
> #(a) Direct Method
> SD_direct <- sqrt(sum((x - mean_x)^2) / length(x))
> SD_direct
[1] 2.627851
> #(b) Assumed Mean Method
> SD_assumed <- sqrt(sum(d^2) / length(x) - (sum(d) /
+                                              length(x))^2)
> SD_assumed
[1] 2.627851
> #(c) Step Deviation Method
> SD_step <- sqrt( (sum(u^2) / length(x)) - (sum(u) /
+                                              length(x))^2 ) * h
> SD_step
[1] 2.627851
>
```

OUTPUT:-



Que 1 (code part):-

```
#Question1:
x <- c(12,15,18,14,16,12,19,15,13,17,15,11,14,18,20,
15,13,16,12,19,14,10,17,15,18)
#MEAN (Three Methods)
#(a) Direct Method
mean_x <- sum(x) / length(x)
```

```
mean_x
#(b) Assumed Mean Method
A <- 15 # assumed mean
d <- x - A
mean_assumed <- A + sum(d) / length(x)
mean_assumed
#(c) Step Deviation Method
h <- 1
u <- (x - A) / h
mean_step <- A + (sum(u) / length(x)) * h
mean_step
#MEDIAN
median_x <- median(x)
median_x
#MODE
freq <- table(x)
mode_x <- as.numeric(names(freq[freq == max(freq)]))
mode_x
#RANGE & COEFFICIENT OF RANGE
range_x <- max(x) - min(x)
range_x
coef_range <- (max(x) - min(x)) / (max(x) + min(x))
coef_range#QUARTILE DEVIATION & COEFFICIENT
Q1 <- as.numeric(quantile(x, 0.25))
Q3 <- as.numeric(quantile(x, 0.75))
QD <- (Q3 - Q1) / 2
QD
coef_QD <- (Q3 - Q1) / (Q3 + Q1)
coef_QD
#MEAN DEVIATION FROM MEAN & COEFFICIENT
MD_mean <- sum(abs(x - mean_x)) / length(x)
MD_mean
coef_MD_mean <- MD_mean / mean_x
coef_MD_mean
```

```
#MEAN DEVIATION FROM MEDIAN & COEFFICIENT
MD_median <- sum(abs(x - median_x)) / length(x)
MD_median
coef_MD_median <- MD_median / median_x
coef_MD_median
#STANDARD DEVIATION (Three Methods)
#(a) Direct Method
SD_direct <- sqrt(sum((x - mean_x)^2) / length(x))
SD_direct
#(b) Assumed Mean Method
SD_assumed <- sqrt(sum(d^2) / length(x) - (sum(d) /
length(x))^2)
SD_assumed
#(c) Step Deviation Method
SD_step <- sqrt( (sum(u^2) / length(x)) - (sum(u) /
length(x))^2 ) * h
SD_step
```

QUE-2

2. A local coffee shop surveyed 40 regular customers to find out how many times they visited the shop in the last week. The results are organized into the frequency table below.

X (No. of Visits) 0 1 2 3 4 5

f (No. of Customers) 4 12 11 7 4 2

CODE:-

DA1.R*

```r
#Question 2:
x <- c(0,1,2,3,4,5)
f <- c(4,12,11,7,4,2)
data <- rep(x, f)
length(data)
#MEAN
#(A)DIRECT METHOD
mean_x <- sum(x * f) / sum(f)
mean_x
#(b) Assumed Mean Method
A <- 2 # assumed mean
d <- x - A
mean_assumed <- A + sum(f * d) / sum(f)
mean_assumed
#(c) Step Deviation Method
h <- 1
u <- (x - A) / h
mean_step <- A + (sum(f * u) / sum(f)) * h
mean_step
#MEDIAN
median_x <- median(data)
median_x
#Mode
mode_x <- x[f == max(f)]
mode_x
#RANGE & COEFFICIENT OF RANGE
range_x <- max(x) - min(x)
range_x
coef_range <- (max(x) - min(x)) / (max(x) + min(x))
coef_range
#QUARTILE DEVIATION (Q.D.) & COEFFICIENT
Q1 <- as.numeric(quantile(data, 0.25))
Q3 <- as.numeric(quantile(data, 0.75))
QD <- (Q3 - Q1) / 2
QD
coef_QD <- (Q3 - Q1) / (Q3 + Q1)
coef_QD
#MEAN DEVIATION ABOUT MEAN & COEFFICIENT
MD_mean <- sum(abs(data - mean_x)) / length(data)
```

59:8 (Top Level) R Script

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CONSOLE:-

Source

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```
> #Question 2:
> x <- c(0,1,2,3,4,5)
> f <- c(4,12,11,7,4,2)
> data <- rep(x, f)
> length(data)
[1] 40
> #MEAN
> #(A)DIRECT METHOD
> mean_x <- sum(x * f) / sum(f)
> mean_x
[1] 2.025
> #(b) Assumed Mean Method
> A <- 2 # assumed mean
> d <- x - A
> mean_assumed <- A + sum(f * d) / sum(f)
> mean_assumed
[1] 2.025
> #(c) Step Deviation Method
> h <- 1
> u <- (x - A) / h
> mean_step <- A + (sum(f * u) / sum(f)) * h
> mean_step
[1] 2.025
> #MEDIAN
> median_x <- median(data)
> median_x
[1] 2
> #Mode
> mode_x <- x[f == max(f)]
> mode_x
[1] 1
> #RANGE & COEFFICIENT OF RANGE
> range_x <- max(x) - min(x)
> range_x
[1] 5
> coef_range <- (max(x) - min(x)) / (max(x) + min(x))
> coef_range
[1] 1
> #QUARTILE DEVIATION (Q.D.) & COEFFICIENT
```

Source

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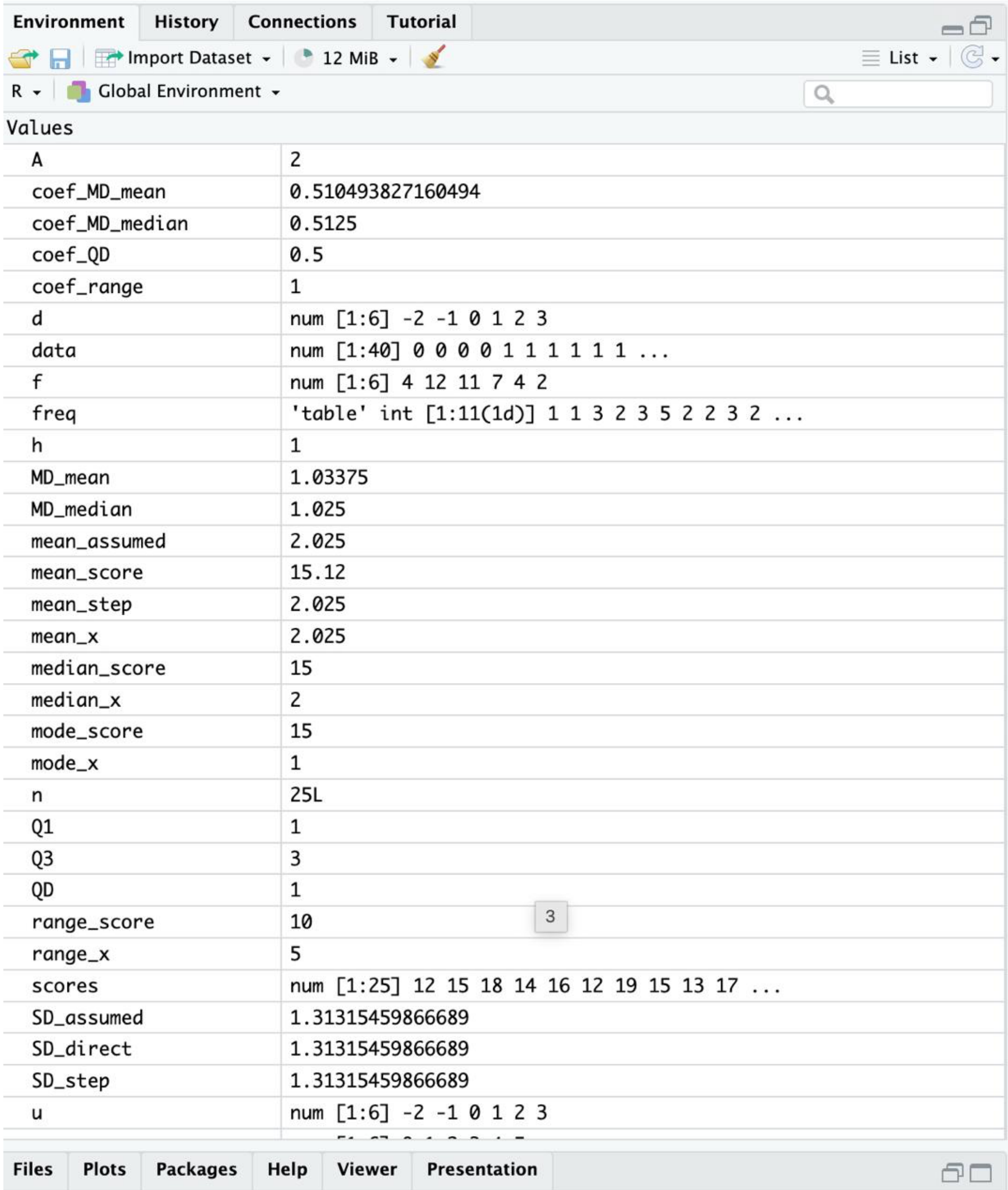
```
> #QUARTILE DEVIATION (Q.D.) & COEFFICIENT
> Q1 <- as.numeric(quantile(data, 0.25))
> Q3 <- as.numeric(quantile(data, 0.75))
> QD <- (Q3 - Q1) / 2
> QD
[1] 1
> coef_QD <- (Q3 - Q1) / (Q3 + Q1)
> coef_QD
[1] 0.5
> #MEAN DEVIATION ABOUT MEAN & COEFFICIENT
> MD_mean <- sum(abs(data - mean_x)) / length(data)
> MD_mean
[1] 1.03375
> coef_MD_mean <- MD_mean / mean_x
> coef_MD_mean
[1] 0.5104938
> #MEAN DEVIATION ABOUT MEDIAN & COEFFICIENT
> MD_median <- sum(abs(data - median_x)) / length(data)
> MD_median
[1] 1.025
> coef_MD_median <- MD_median / median_x
> coef_MD_median
[1] 0.5125
> #STANDARD DEVIATION
> #(a) Direct Method
> SD_direct <- sqrt(sum(f * (x - mean_x)^2) / sum(f))
> SD_direct
[1] 1.313155
> #(b) Assumed Mean Method
> SD_assumed <- sqrt(sum(f * d^2) / sum(f) - (sum(f * d) /
+                                                  sum(f))^2)
> SD_assumed
[1] 1.313155
> #(c) Step Deviation Method
> SD_step <- sqrt(sum(f * u^2) / sum(f) - (sum(f * u) /
+                                               sum(f))^2) * h
> SD_step
[1] 1.313155
> 
```

OUTPUT:-



Values	
A	2
coef_MD_mean	0.510493827160494
coef_MD_median	0.5125
coef_QD	0.5
coef_range	1
d	num [1:6] -2 -1 0 1 2 3
data	num [1:40] 0 0 0 0 1 1 1 1 1 1 ...
f	num [1:6] 4 12 11 7 4 2
freq	'table' int [1:11(1d)] 1 1 3 2 3 5 2 2 3 2 ...
h	1
MD_mean	1.03375
MD_median	1.025
mean_assumed	2.025
mean_score	15.12
mean_step	2.025
mean_x	2.025
median_score	15
median_x	2
mode_score	15
mode_x	1
n	25L
Q1	1
Q3	3
QD	1
range_score	10
range_x	5
scores	num [1:25] 12 15 18 14 16 12 19 15 13 17 ...
SD_assumed	1.31315459866689
SD_direct	1.31315459866689
SD_step	1.31315459866689
u	num [1:6] -2 -1 0 1 2 3

Que –2(code part):-

```
#Question 2:
x <- c(0,1,2,3,4,5)
f <- c(4,12,11,7,4,2)
data <- rep(x, f)
length(data)
#MEAN
#(A)DIRECT METHOD
```

```
mean_x <- sum(x * f) / sum(f)
mean_x
#(b) Assumed Mean Method
A <- 2 # assumed mean
d <- x - A
mean_assumed <- A + sum(f * d) / sum(f)
mean_assumed
#(c) Step Deviation Method
h <- 1
u <- (x - A) / h
mean_step <- A + (sum(f * u) / sum(f)) * h
mean_step
#MEDIAN
median_x <- median(data)
median_x
#Mode
mode_x <- x[f == max(f)]
mode_x
#RANGE & COEFFICIENT OF RANGE
range_x <- max(x) - min(x)
range_xcoef_range <- (max(x) - min(x)) / (max(x) + min(x))
coef_range
#QUARTILE DEVIATION (Q.D.) & COEFFICIENT
Q1 <- as.numeric(quantile(data, 0.25))
Q3 <- as.numeric(quantile(data, 0.75))
QD <- (Q3 - Q1) / 2
QD
coef_QD <- (Q3 - Q1) / (Q3 + Q1)
coef_QD
#MEAN DEVIATION ABOUT MEAN & COEFFICIENT
MD_mean <- sum(abs(data - mean_x)) / length(data)
MD_mean
coef_MD_mean <- MD_mean / mean_x
coef_MD_mean
```

```
#MEAN DEVIATION ABOUT MEDIAN & COEFFICIENT
MD_median <- sum(abs(data - median_x)) / length(data)
MD_median
coef_MD_median <- MD_median / median_x
coef_MD_median
#STANDARD DEVIATION
#(a) Direct Method
SD_direct <- sqrt(sum(f * (x - mean_x)^2) / sum(f))
SD_direct
#(b) Assumed Mean Method
SD_assumed <- sqrt(sum(f * d^2) / sum(f) - (sum(f * d) /
sum(f))^2)
SD_assumed
#(c) Step Deviation Method
SD_step <- sqrt(sum(f * u^2) / sum(f) - (sum(f * u) /
sum(f))^2) * h
SD_step
```

QUESTION 3:-

3. A housing society recorded the monthly electricity consumption (in kilowatt-hours, kWh) of 150 households to analyse energy usage patterns. The data is summarized in the table below:

Consumption (kWh)	100-150	150-200	200-250	250-300	300-350	350-400	400-450
No. of Households (f)	12	28	45	35	18	10	2

CODE:-

DA1.R*

```
#question 3:
# Class intervals
lower <- c(100,150,200,250,300,350,400)
upper <- c(150,200,250,300,350,400,450)
# Frequencies
f <- c(12,28,45,35,18,10,2)
# Class mid-points
mid <- (lower + upper) / 2
sum(f) # total households
#Mean
#(a) Direct Method
mean_x <- sum(mid * f) / sum(f)
mean_x
#(b) Assumed Mean Method
A <- 225 # assumed mean
d <- mid - A
mean_assumed <- A + sum(f * d) / sum(f)
mean_assumed
#(c) Step Deviation Method
h <- 50
u <- (mid - A) / h
mean_step <- A + (sum(f * u) / sum(f)) * h
mean_step
#Median
cf <- cumsum(f)
cf
N <- sum(f)
median_class <- which(cf >= N/2)[1]
l <- lower[median_class]
c_prev <- ifelse(median_class == 1, 0, cf[median_class -
                                          1])
f_m <- f[median_class]
median_x <- l + ((N/2 - c_prev) / f_m) * h
median_x
#Mode
modal_class <- which.max(f)
l1 <- lower[modal_class]
f1 <- f[modal_class]
f0 <- ifelse(modal_class == 1, 0, f[modal_class - 1])
```

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CONSOLE:-

Source

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```
> #Mode
> mode_x <- x[f == max(f)]
> mode_x
[1] 1
> #RANGE & COEFFICIENT OF RANGE
> range_x <- max(x) - min(x)
> range_x
[1] 5
> coef_range <- (max(x) - min(x)) / (max(x) + min(x))
> coef_range
[1] 1
> #QUARTILE DEVIATION (Q.D.) & COEFFICIENT
> Q1 <- as.numeric(quantile(data, 0.25))
> Q3 <- as.numeric(quantile(data, 0.75))
> QD <- (Q3 - Q1) / 2
> QD
[1] 1
> coef_QD <- (Q3 - Q1) / (Q3 + Q1)
> coef_QD
[1] 0.5
> #MEAN DEVIATION ABOUT MEAN & COEFFICIENT
> MD_mean <- sum(abs(data - mean_x)) / length(data)
> MD_mean
[1] 1.03375
> coef_MD_mean <- MD_mean / mean_x
> coef_MD_mean
[1] 0.5104938
> #MEAN DEVIATION ABOUT MEDIAN & COEFFICIENT
> MD_median <- sum(abs(data - median_x)) / length(data)
> MD_median
[1] 1.025
> coef_MD_median <- MD_median / median_x
> coef_MD_median
[1] 0.5125
> #STANDARD DEVIATION
> #(a) Direct Method
> SD_direct <- sqrt(sum(f * (x - mean_x)^2) / sum(f))
> SD_direct
[1] 1.313155
> #(b) Assumed Mean Method
```

Source

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```
> l3 <- lower[Q3_class]
> c3 <- cf[Q3_class - 1]
> f3 <- f[Q3_class]
> Q3 <- l3 + ((3*N/4 - c3) / f3) * h
> QD <- (Q3 - Q1) / 2
> QD
[1] 46.875
> coef_QD <- (Q3 - Q1) / (Q3 + Q1)
> coef_QD
[1] 0.1933702
> #MEAN DEVIATION ABOUT MEAN & COEFFICIENT
> MD_mean <- sum(f * abs(mid - mean_x)) / sum(f)
> MD_mean
[1] 56.2
> coef_MD_mean <- MD_mean / mean_x
> coef_MD_mean
[1] 0.2303279
> #MEAN DEVIATION ABOUT MEDIAN & COEFFICIEN
> MD_median <- sum(f * abs(mid - median_x)) / sum(f)
> MD_median
[1] 55.51852
> coef_MD_median <- MD_median / median_x
> coef_MD_median
[1] 0.2324031
> #STANDARD DEVIATION
> #(a) Direct Method
> SD_direct <- sqrt(sum(f * (mid - mean_x)^2) / sum(f))
> SD_direct
[1] 68.71924
> #(b) Assumed Mean Method
> SD_assumed <- sqrt(sum(f * d^2) / sum(f) - (sum(f * d) /
+                                                 sum(f))^2)
> SD_assumed
[1] 68.71924
> #(c) Step Deviation Method
> SD_step <- sqrt(sum(f * u^2) / sum(f) - (sum(f * u) /
+                                               sum(f))^2) * h
> SD_step
[1] 68.71924
```

OUTPUT:-

Environment | History | Connections | Tutorial

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lower	num [1:7] 100 150 200 250 300 350 400
MD_mean	56.2
MD_median	55.5185185185185
mean_assumed	244
mean_score	15.12
mean_step	244
mean_x	244
median_class	3L
median_score	15
median_x	238.888888888889
mid	num [1:7] 125 175 225 275 325 375 425
modal_class	3L
mode_score	15
mode_x	231.481481481481
n	25L
N	150
Q1	195.535714285714
Q1_class	2L
Q3	289.285714285714
Q3_class	4L
QD	46.875
range_score	10
range_x	range_score (numeric , 56 bytes)
scores	num [1:25] 12 15 18 14 16 12 19 15 13 17 ...
SD_assumed	68.7192355409556
SD_direct	68.7192355409556
SD_step	68.7192355409556
u	num [1:7] -2 -1 0 1 2 3 4
upper	num [1:7] 150 200 250 300 350 400 450
x	num [1:6] 0 1 2 3 4 5
Functions	
mode_function	function (x)

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Que-3(code part):-

```
#question 3:
# Class intervals
lower <- c(100,150,200,250,300,350,400)
upper <- c(150,200,250,300,350,400,450)
# Frequencies
f <- c(12,28,45,35,18,10,2)
# Class mid-points
```

```
mid <- (lower + upper) / 2
sum(f) # total households
#Mean
#(a) Direct Method
mean_x <- sum(mid * f) / sum(f)
mean_x
#(b) Assumed Mean Method
A <- 225 # assumed mean
d <- mid - A
mean_assumed <- A + sum(f * d) / sum(f)
mean_assumed
#(c) Step Deviation Method
h <- 50
u <- (mid - A) / h
mean_step <- A + (sum(f * u) / sum(f)) * h
mean_step
#Median
cf <- cumsum(f)
cf
N <- sum(f)
median_class <- which(cf >= N/2)[1]
l <- lower[median_class]c_prev <- ifelse(median_class == 1, 0,
cf[median_class -
1])
f_m <- f[median_class]
median_x <- l + ((N/2 - c_prev) / f_m) * h
median_x
#Mode
modal_class <- which.max(f)
l1 <- lower[modal_class]
f1 <- f[modal_class]
f0 <- ifelse(modal_class == 1, 0, f[modal_class - 1])
f2 <- ifelse(modal_class == length(f), 0, f[modal_class +
1])
```

```
mode_x <- l1 + ((f1 - f0) / (2*f1 - f0 - f2)) * h
mode_x
#RANGE & COEFFICIENT OF RANGE
range_x <- max(upper) - min(lower)
range_x
coef_range <- (max(upper) - min(lower)) / (max(upper) +
min(lower))
coef_range
#QUARTILE DEVIATION (Q.D.) & COEFFICIENT
# Q1
Q1_class <- which(cf >= N/4)[1]
l1 <- lower[Q1_class]
c1 <- ifelse(Q1_class == 1, 0, cf[Q1_class - 1])
f1 <- f[Q1_class]
Q1 <- l1 + ((N/4 - c1) / f1) * h
# Q3
Q3_class <- which(cf >= 3*N/4)[1]
l3 <- lower[Q3_class]
c3 <- cf[Q3_class - 1]f3 <- f[Q3_class]
Q3 <- l3 + ((3*N/4 - c3) / f3) * h
QD <- (Q3 - Q1) / 2
QD
coef_QD <- (Q3 - Q1) / (Q3 + Q1)
coef_QD
#MEAN DEVIATION ABOUT MEAN & COEFFICIENT
MD_mean <- sum(f * abs(mid - mean_x)) / sum(f)
MD_mean
coef_MD_mean <- MD_mean / mean_x
coef_MD_mean
#MEAN DEVIATION ABOUT MEDIAN & COEFFICIEN
MD_median <- sum(f * abs(mid - median_x)) / sum(f)
MD_median
coef_MD_median <- MD_median / median_x
coef_MD_median
```

```
#STANDARD DEVIATION
#(a) Direct Method
SD_direct <- sqrt(sum(f * (mid - mean_x)^2) / sum(f))
SD_direct
#(b) Assumed Mean Method
SD_assumed <- sqrt(sum(f * d^2) / sum(f) - (sum(f * d) /
sum(f))^2)
SD_assumed
#(c) Step Deviation Method
SD_step <- sqrt(sum(f * u^2) / sum(f) - (sum(f * u) /
sum(f))^2) * h
SD_step
```